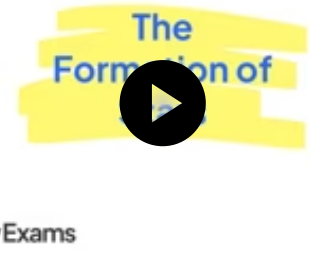
Star Formation

Exam code: 0625 & 0972

Download PDF

Written by: Katie M
Reviewed by: Caroline Carroll
Updated on 18 June 2025



Did this video help you? Yes No

Help me understand

Star formation

Extended tier only

- Stars go through a sequence of evolutionary stages, known as the **life cycle** of a star
- All stars follow the same initial stages:

nebula → protostar → stable star

Nebula

- Stars form from a giant interstellar **cloud of gas and dust** called a nebula

Protostar

- The **gravitational attraction** within a nebula pulls the particles **closer together** until a hot ball of gas forms, known as a **protostar**
- As the particles are pulled closer together, the **density** of the protostar **increases**
- This results in **more frequent collisions** between the particles, which causes the **temperature** to **increase**

Stable star

- Once the protostar becomes hot enough, **nuclear fusion** reactions occur within its core

Stuck on something?

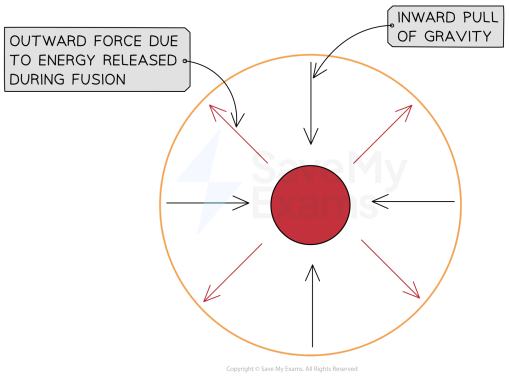
Ask questions and get help with exactly what you're revising.

Got it

keeps the core of the star hot

- Once a star initiates fusion, it is known as a **stable star**
- During this stage, the star is in **equilibrium** as the forces acting on it are balanced
- Gravitational forces act **inwards**
  - This is an attractive force which pulls the outer layers inwards
- Thermal pressure acts **outwards**
  - This is exerted by the expanding hot gases inside the star as energy is released during fusion

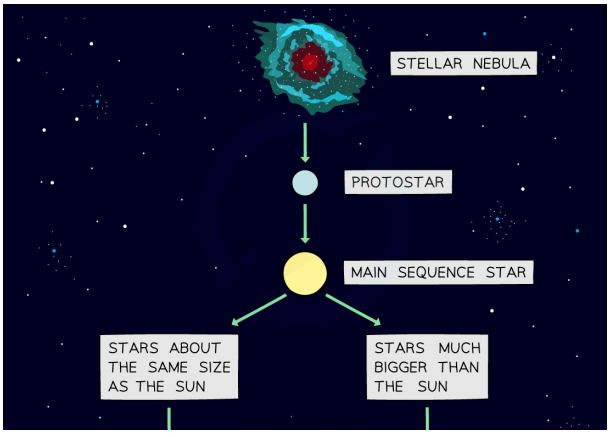
Balanced forces in a stable star

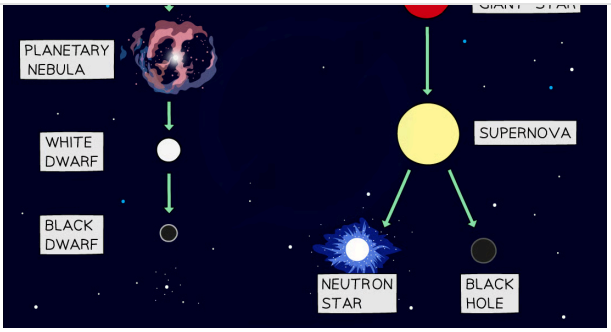


The outward and inward forces within a star are in equilibrium. The centre red circle represents the star's core and the orange circle represents the star's outer layers

- Once a stable star is formed, the final stages of its life cycle depend on its **mass**
- The different life cycles are shown below

Summary of the life cycles of stars





Flow diagram showing the life cycle of a star which is the same size as the Sun (solar mass) and the lifecycle of a star which is much more massive than the Sun

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder →

Test yourself Flashcards

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub

Scholarship

Join

Dashboard

Overview

Brand ambassador

Profile

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.